

Allison (Amy) O'Connell

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EDUCATION

Ph.D., Computer Science. 2021–2026
University of Southern California (USC) GPA: 4.0
Dissertation: Toward Iterative Development of Modular, Low-Cost, Open-Source Robots for In-the-Wild Human-Robot Interaction Studies,
Advisor: Maja Matarić

B.A., Computer Science. 2016–2020
Vassar College GPA: 3.77

PROFESSIONAL

IT Intern - Intelligent Solutions Group, John Deere. *Supervisor:* Siddharth Thakur May 2023–Aug 2023
Visiting Researcher, University of South Carolina Autonomous Field Robotics Lab *Advisor:* Jason O'Kane, Ioannis Rekleitis May 2019–Aug 2019

TEACHING

Teaching Assistant, CSCI 103–*Introduction to Programming*. University of Southern California. Developed exams, advised project teams, hosted office hours, and directed and managed lab sections in Spring 2025.

Teaching Assistant, CSCI 201–*Principles of Software Development*. University of Southern California. Directed and managed lab sections, hosted office hours, and developed exams in Summer 2024 and Fall 2025.

Student Teaching Assistant, CS 145–*Mathematical Foundations of Computer Science*. Vassar College. Managed lab sections and hosted office hours in Spring 2019.

Student Teaching Assistant, CS 101–*Problem-Solving and Abstraction*. Vassar College. Managed lab sections and hosted office hours in Fall 2018 and Fall 2019.

Student Teaching Assistant, CS 101–CS 241: *Analysis of Algorithms*. Vassar College. Hosted office hours in-person and via Zoom in Spring 2020.

RESEARCH SUPERVISION

Kevin Zhang,	Aug 2021–May 2022
Bailey Cislowski,	Sept 2022–May 2024
Andrew Le,	March 2022–Present
Ellen Ko,	March 2022–May 2024
Nikki Yaminrafie,	March 2022–May 2024
Jennifer Ayissi,	Sept 2022–Feb 2025
Caroline Kenney,	Feb 2023–Present
Amani Yohanes,	September 2023–Present
Maya Reetz,	December 2023–Present
Kai Herman,	June 2024–August 2024
Sanya Khattar,	August 2024–January 2026
Alex Severinghaus,	August 2024–December 2024
John-Paul Yoder,	September 2025–Present
Salinaa Naik,	January 2026–Present
Leyaa George,	January 2026–Present

USC Viterbi Summer Undergraduate Research Experience (SURE)
Stephania Castro, June 2024–August 2024

External NSF REU Summer Fellows

Vi Vo,
Ashveen Banga,

June 2022–Aug 2022
June 2023–August 2023

Summer High School Intensive in Next-Generation Engineering (SHINE)

Claire Leshner,
Dru Reed,
CJ Phelan,
Rami Seid,

June 2022–Aug 2022
June 2022–Aug 2022
June 2024–Aug 2024
June 2024–Aug 2024

**HONORS AND
AWARDS***2026 IEEE/ACM Conference on Human-Robot Interaction*

HRI Pioneers

March 2026

Awarded to the top 23% of HRI Pioneers applicants.

2026 ACM conference presenting the latest research and innovations in tangible, embedded, and embodied interaction

Graduate Student Consortium

March 2026

One of eight awardees to attend a mentorship session with established academic researchers in the TEI community.

CO-AUTHORED
AWARDED GRANTS

NSF/NIH Smart Health and Biomedical Research in the Era of Artificial Intelligence and Advanced Data Science (\$1,200,000 total) Aug 2024–Aug 2028
“Personalized AI-Driven Models for Supporting User Engagement and Adherence in Health Interventions: Validation in Cognitive Behavioral Therapy for Anxiety,” PI: Maja Matarić, Co-PIs: Stefanos Nikolaidis, Bruna Martins-Klein.
Motivated problem through the psychological framework of self-determination theory, provided background on self-determination theory, described mathematical formulation for learning user preferences from explicit user feedback, designed all user studies (co-design sessions and randomized controlled trials), and identified outcome instruments and measurements.

PUBLICATIONS

Journal Publications

[J3] RA-M '25 Massimiliano Nigro, **Amy O’Connell**, Thomas Groechel, Anna-Maria Velentza, and Maja J. Matarić. “An Interactive Augmented Reality Interface for Personalized Proxemics Modeling: Comfort and Human–Robot Interactions.” *IEEE Robotics & Automation Magazine* 32, no. 3 (2025): 125-133. DOI: [10.1109/MRA.2024.3415108](https://doi.org/10.1109/MRA.2024.3415108)
(IF: 7.1)

Conference Publications

[C12] CHI '25 Maria R. Lima, **Amy O’Connell**, Feiyang Zhou, Alethea Nagahara, Avni Hulyalkar, Anura Deshpande, Jesse Thomason, Ravi Vaidyanathan, and Maja Matarić. “Promoting Cognitive Health in Elder Care with Large Language Model-Powered Socially Assistive Robots.” *ACM CHI Conference on Human Factors in Computing Systems*, Yokohama, Japan, April 26–May 1, 2025.

[C11] HRI '24 **Amy O’Connell**, Ashveen Banga, Jennifer Ayissi, Nikki Yaminrafie, Ellen Ko, Andrew Le, Bailey Cislowski, and Maja Matarić. “Design and Evaluation of a Socially Assistive Robot School-work Companion for College Students with ADHD.” *ACM/IEEE Conference on Human-Robot Interaction (HRI)*, Boulder, CO, March 10–13, 2024.
(25% Acceptance Rate)

- [C9] AAAI '24 Zhonghao Shi, **Amy O'Connell**, Zongjian Li, Siqi Liu, Jennifer Ayissi, Guy Hoffman, Mohammad Soleymani, and Maja J Matarić. "Build Your Own Robot Friend: An Open-Source Learning Module for Accessible and Engaging AI Education.", *AAAI Conference on Artificial Intelligence Symposium on Educational Advances in Artificial Intelligence (EAAI)*, Vancouver, Canada, Feb 20–27, 2024. DOI: [10.1145/3568162.3576987](https://doi.org/10.1145/3568162.3576987)
- [C3] HRI '23 Zhonghao Shi, Han Chen, Anna-Maria Velentza, Siqi Liu, Nathaniel Dennler, **Amy O'Connell**, and Maja Mataric. "Evaluating and Personalizing User-Perceived Quality of Text-to-Speech Voices for Delivering Mindfulness Meditation with Different Physical Embodiments", *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, Stockholm, Sweden, Mar 13–16, 2023. DOI: [10.1145/3568162.3576987](https://doi.org/10.1145/3568162.3576987) (24% Acceptance Rate)
- [C7] RO-MAN '22 Thomas R. Groechel, **Amy O'Connell**, Massimiliano Nigro, Maja J. Matarić. "Reimagining rviz: Multidimensional augmented reality robot signal design.", 31st *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, Naples, Italy, Aug 29–Sept 2, 2022. DOI: [10.1109/RO-MAN53752.2022.9900692](https://doi.org/10.1109/RO-MAN53752.2022.9900692)

Refereed Workshop Papers

- [WS1] HRI '24 **Nathaniel Dennler**, Amy O'Connell, Stefanos Nikolaidis, Maja Matarić. "Robot Costume Design: Identity as a Narrative", *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, at the 3rd Workshop on Robo-Identity, Boulder, CO, Mar 11-15, 2024.

INVITED TALKS

"Designing Accessible Platforms for Robotics Research and Education" June 2024
invited speaker for *MIRRORLab Summer Speakers* at **Colorado School of Mines**. Presented Virtually.

"Designing Accessible Platforms for Robotics Research and Education" July 2024
invited lecture for *USC Summer Robotics Seminar Series* at **University of Southern California**. Los Angeles, CA.

SERVICE

Workshop Organizing Committee

Co-Lead Organizer (with Zhonghao Shi, Nathan Dennler, Leigh Levinson, Tiantian Feng, Xuan Shi, Nicholas Georgiou, Jieyu Zhao, Shrikanth Narayanan, Brian Scassellati, Selma Šabanović, Maja Matarić), *AAAI Spring Symposium* on Child-AI Interaction in the Era of Foundation Models, Burlingame, CA, March 31st–April 2, 2025.

Service Roles

Journal Reviewer

IEEE Transactions on Affective Computing, Autonomous Robots, Science Robotics, and Public Library of Science ONE (PLOS ONE), Frontiers in Robotics, Transactions on Human-Robot Interaction (T-HRI), Autonomous Robotics

Conference Reviewer

Human-Robot Interaction (HRI), International Conference on Robot and Human Interactive Communication (RO-MAN), International Conference on Social Robotics (ICSR), International Society of the Learning Sciences (ISLS), Haptics Symposium

Lead Organizer

September 2021–Present

Women in US Academic Research in Robotics List

1. Maintain a public list of Women in US Academic Research in Robotics.
2. Solicit new entries, determine eligibility and add new entries to the list.

Tour Director

September 2021–Present

USC Interaction Lab

1. Main point of contact for local school and community groups seeking tours of the USC Interaction Lab.
2. Schedule visits and connect tour groups with a lab member tour guide.
3. Communicate with tour groups on day of visit.
4. Lead over 20 tours of the Interaction Lab.

Chair

August 2018–May 2020

Computer Science Majors Committee

1. Supported Vassar Computer Science students and acted as a liaison between students and professors.
2. Organized department events, student panels, and informal gatherings.
3. Helped hire new professors by interviewing candidates, attending guest lectures, and providing feedback.